Continuum limit

$$\omega(k) = 2J\sqrt{1+g^2-2g\cos(ka)} \underset{\substack{\text{expand}\\ \text{near } k=0}}{=\!=} 2J\sqrt{1+g^2-2g\left(1-\tfrac{1}{2}(ka)^2\right)} =$$

$$= 2J\sqrt{(1-g)^2 + g a^2 k^2} = 2Ja\sqrt{\frac{(g-g_c)^2}{a^2} + k^2}$$

$g_c = 1$; "mass" $m^2 = \frac{(g-g_c)^2}{a^2}$

$$\omega(k) \cong 2Ja\; m\left(1+\frac{k^2}{2m^2}\right) = 2Ja\left(m + \frac{k^2}{2m}\right)$$

for $k \ll m$

$$\omega(k) \approx 2Jak = ck \quad \text{for } k \gg m$$

At the quantum transition $g = g_c$, $m = 0$, diverging length $\xi \approx \frac{a}{|g-g_c|}$ is correlation length.

Expand Hamiltonian in low-energy modes

$$c_x = \int_{-\pi}^{\pi} \frac{dk}{2\pi} e^{ikx} c_k \longrightarrow \int_{-\Lambda}^{\Lambda} \frac{dk}{2\pi} e^{ikx} c_k = \psi(x)$$

$$\{\psi(x), \psi^\dagger(x')\} = \int \frac{dk_1}{2\pi}\frac{dk_2}{2\pi} e^{ik_1 x - ik_2 x'} \underbrace{\{c_{k_1}, c^\dagger_{k_2}\}}_{\delta_{k_1,k_2}} =$$

$$= \int \frac{dk}{2\pi} e^{ik(x-x')} = \delta(x-x')$$

Then $\sum_k (1-g\cos ka)\, c^\dagger_k c_k \approx (1-g)\sum_k c^\dagger_k c_k$

let $c_k = \frac{1}{\sqrt{L}} \int dx\, e^{-ikx} \psi(x)$

$$\sum_k c_k^+ c_k = L \int \frac{dk}{2\pi} \frac{1}{L} \int dx\, dx'\, e^{ikx - ikx'} \psi^+(x) \psi(x) =$$

$$= \int dx\, dx'\, \delta(x-x')\, \psi^+(x) \psi(x) = \int dx\, \psi^+(x) \psi(x).$$

$$J \sum_k i \sin(ka)\, c_k^+ c_{-k}^+ = JL \int \frac{dk}{2\pi} \underbrace{i \sin(ka)}_{ika} \frac{1}{L} \int dx\, dx'\, e^{ikx - ikx'}$$

$$\cdot\, \psi^+(x) \psi^+(x')$$

$$= J \int \frac{dk}{2\pi} \int dx\, dx'\, (ia)\, i \frac{\partial}{\partial x'}\, e^{ikx - ikx'}\, \psi^+(x) \psi^+(x') =$$

$$= J \int \frac{dk}{2\pi} \int dx\, dx'\, (-a)\, e^{ik(x-x')}\, \psi^+(x)\, \partial_{x'}\, \psi^+(x') =$$

$$= (-Ja) \int dx\, \psi^+(x) \frac{\partial}{\partial x} \psi^+(x)$$

$$\Rightarrow J \sum_k -ig \sin(ka)\, c_k^+ c_{-k}^+ = \underbrace{gJa}_{g_c \cdot \frac{c}{2}} \int dx\, \psi^+(x)\, \partial_x \psi^+(x)$$

$$\Rightarrow H = \int dx\, \frac{c}{2} \left(\psi^+ \partial_x \psi^+ - \psi \partial_x \psi \right) + \Delta \psi^+ \psi$$

$$\Delta = 2J|1-g|$$

$$Z = \mathrm{Tr}\, e^{-\beta \hat{H}} = \int D\psi D\psi^+\, e^{-\int dx \int_0^\beta d\tau \left(\psi^+ \partial_\tau \psi + H \right)}$$

Path integral for fermions is written in terms of Grassmann variables ψ (R. Shankar "Principles of Quantum Mechanics" p. 640-653)

Critical theory $\Delta = 0$ $(g = g_c = 1)$

Require scale invariance of S with $\Delta = 0$

$$\int dx\, d\tau \left\{ \psi^+ \frac{\partial \psi}{\partial \tau} + \frac{c}{2}\left(\psi^+ \frac{\partial \psi^+}{\partial x} - \psi \frac{\partial \psi}{\partial x} \right) \right\}$$

Dimensions $[x] = L$, $[\tau] = L^z$

$$\Rightarrow \left.\begin{aligned} &\int dx\, d\tau\ \psi^+ \frac{\partial \psi}{\partial \tau} \sim L\,[\psi]^2 = 1 \\ &\int dx\, d\tau\ \psi \frac{\partial \psi}{\partial x} \sim L^z\,[\psi]^2 = 1 \end{aligned}\right\} \quad \begin{aligned} &[\psi] = \frac{1}{L^{1/2}} \\ &z = 1 \end{aligned}$$

$[\psi] = L^{-1/2}$ means that

$$\langle \psi(x)\, \psi^+(0) \rangle \sim \frac{1}{x}$$

Now dimension of

$$\int dx\, d\tau\ \Delta\, \psi^+ \psi = L^2 [\psi]^2 = L$$

that is grows as length grows and controls long distance properties $\Delta L_0 \sim 1 \Rightarrow \xi = L_0 = \frac{1}{\Delta}$.

So Δ is analog of $t = \frac{T - T_c}{T_c}$ as it represents <u>relevant</u> direction.

All other terms that can be added are <u>irrelevant</u>

$$\int dx\, d\tau\ \psi^+ \partial_x^2 \psi \sim L^2 \frac{1}{L} \frac{1}{L^2} \sim \frac{1}{L}$$

$$\int dx\, d\tau\ (\psi^+ \partial_x \psi)^+ (\psi \partial_x \psi) \sim L^2 (L^{-1/2})^4 \frac{1}{L^2} \sim \frac{1}{L^2}$$

Emergence of Dirac equation

$$H = iJ \sum_n \left(g \bar{\chi}_n \chi_{n+1} - \bar{\chi}_n \chi_n \right)$$

$$i \partial_t \mathcal{O} = [H, \mathcal{O}]$$

$$i \partial_t \chi_m = iJ \sum_n \left[g \bar{\chi}_n \chi_{n+1} - \bar{\chi}_n \chi_n, \chi_m \right] =$$

$$= iJ \Big\{ g \left(\bar{\chi}_n \chi_{n+1} \chi_m - \chi_m \bar{\chi}_n \chi_{n+1} \right)$$

$$- \left(\bar{\chi}_n \chi_n \chi_m - \chi_m \bar{\chi}_n \chi_n \right) \Big\} =$$

$$= iJ \sum_n 2g \bar{\chi}_n \delta_{m,n+1} - 2 \bar{\chi}_n \delta_{m,n} =$$

$$= iJ\, 2 \left(g \bar{\chi}(m-1) - \bar{\chi}(m) \right)$$

$$i \partial_t \bar{\chi}_m = iJ \sum_n g \left(\bar{\chi}_n \chi_{n+1} \bar{\chi}_m - \bar{\chi}_m \bar{\chi}_n \chi_{n+1} \right) - \left(\bar{\chi}_n \chi_n \bar{\chi}_m - \right.$$

$$\left. - \bar{\chi}_m \bar{\chi}_n \chi_n \right) = iJ \sum_n \left(-2g \chi_{n+1} \delta_{n,m} + 2 \chi_n \delta_{n,m} \right)$$

$$= 2iJ \left(-g \chi(m+1) + \chi(m) \right)$$

Continuum approximation: $\chi(m+1) \approx \chi_m + a \partial_x \chi_m + \dots$

$$i \partial_t \chi_m = 2iJ \left[(g-1) \bar{\chi}_m - a g \partial_x \bar{\chi}_m \right]$$

$$i \partial_t \bar{\chi}_m = 2iJ \left[(1-g) \chi_m - a g \partial_x \chi_m \right]$$

let $\chi_\pm = \frac{1}{2} (\chi \pm \bar{\chi}) \Rightarrow \begin{cases} \chi = \chi_+ + \chi_- \\ \bar{\chi} = \chi_+ - \chi_- \end{cases}$

$$\Rightarrow i \partial_t \chi_+ (x) = 2iJ \left[-(g-1) \chi_- (x) - a g \partial_x \chi_+ \right]$$

$$i \partial_t \chi_- (x) = 2iJ \left[(g-1) \chi_+ + a g \partial_x \chi_- \right]$$

let $m = \frac{g-1}{a}$, $v = aJ$, and set $g=1$ in $(a g \partial_x \chi_\pm)$:

$$\Rightarrow \begin{cases} \frac{1}{v}\partial_t \chi_+ = -\partial_x \chi_+ - m\chi_- \\ \frac{1}{v}\partial_t \chi_- = +\partial_x \chi_- + m\chi_+ \end{cases}$$

At $g=1$ $(m=0)$ $\chi_\pm$ are chiral fermions

$$\frac{1}{v}\frac{\partial}{\partial t}\chi_\pm \pm \frac{\partial}{\partial x}\chi_\pm = 0$$

More generally,

$$\begin{pmatrix} m & -\frac{1}{v}\partial_t + \partial_x \\ \frac{1}{v}\partial_t + \partial_x & m \end{pmatrix}\begin{pmatrix} \chi_+ \\ \chi_- \end{pmatrix} = 0$$

is actually two-dimensional Dirac equation.

$$\gamma^0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$(\gamma^0)^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\hat{1}$$

$$(\gamma^1)^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{1}$$

$$\gamma^0\gamma^1 + \gamma^1\gamma^0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 0.$$

$$\left[\gamma^0 \frac{\partial}{\partial x^0} + \gamma^1 \frac{\partial}{\partial x^1} + m\right]\chi = 0$$

$$\gamma^\mu \frac{\partial}{\partial x^\mu}\chi + m\chi = 0.$$